



UNIVERSITÄT BONN

Juergen Gall

Linear Filtering Computer Vision

MA-INF 2201 / MA-MOROB-M04
WS25/26

Organization

- Lecturer:
 - Prof. Dr. Juergen Gall
 - https://pages.iai.uni-bonn.de/gall_juergen
- Teaching:

Prof. Dr. Juergen Gall

Research Interests

Job Offers/Theses

Publications

Software

Data

Projects

Conferences

Teaching

Teaching

PhD Seminar - Graphics, Vision, Audio for Intelligent Systems, SS13, WS13/14, SS14, WS14/15, SS15, WS15/16, SS16, WS16/17, SS17, WS17/18, SS18, WS18/19, SS19, WS19/20, SS20, WS20/21, SS21, WS21/22, SS22, WS22/23, SS23, WS23/24, SS24, WS24/25, SS25, WS25/26

BA-INF 062 - Begleitsseminar zur Bachelorarbeit: Computer Vision, SS14, WS14/15, SS15, WS15/16, SS16, WS16/17, SS17, WS17/18, SS18, WS18/19, SS19, WS19/20, SS20, WS20/21, SS21, WS21/22, SS22, WS22/23, SS23, WS23/24, SS24, WS24/25, SS25, WS25/26

MA-INF 2201/MA-MOROB-M04 - Computer Vision, SS13/14, WS13/14, SS15/16, WS16/17, WS17/18, WS18/19, WS19/20, WS21/22, WS22/23, WS23/24, WS24/25, WS25/26

MA-INF 2213 - Advanced Computer Vision (Computer Vision II), SS14, SS15, SS16, SS17, SS18, SS19, SS20, SS21, SS22, SS23, SS24, SS25

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Organization

- Slides:
 - https://pages.iai.uni-bonn.de/gall_juergen/teaching/Lectures/cv25.html

MA-INF 2201/MA-MOROB-M04 - Computer Vision

4L + 2E, WS25/26

Lecturer

Juergen Gall

Degree programs

The lecture is offered for the Master of Science degrees: Computer Science (MA-INF 2201) and Mobile Robotics (MA-MOROB-M04).

Content

Treatative: Linear filters, Edges, Derivatives, Hough Transform, Segmentation, Graph Cuts, Mean Shift, Active Contours, Level Sets, MRFs, Expectation Maximization, Temporal Filtering, Active Appearance Models, Shapes, Optical Flow, 2D Tracking, Cameras, 2D/3D Features, Stereo, 3D Reconstruction, 3D Pose Estimation, A Deformable Meshes, RGBD Vision

Material

The slides and recordings are available at [Slides/Recordings](#).

Prerequisites

Basic knowledge of linear algebra, analysis, probability theory, Python programming

Schedule

Tuesday, 10:15-11:45, HS XVI, Nulallee 17

Friday, 10:15-11:45, HS XVI, Nulallee 17

Start: Tuesday, 14.10.

Organization

- Slides:



- Password:** MA-INF2201

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Organization

- Exercise:
 - Room 2.025
Institute of Computer Science
Friedrich-Hirzebruch-Allee 8
 - Start: 22.10.
 - Wednesday, 12:00-13:00
 - Wednesday, 13:00-14:00
 - Wednesday, 14:00-15:00
 - Wednesday, 15:00-16:00
 - You will get a slot assignment soon**

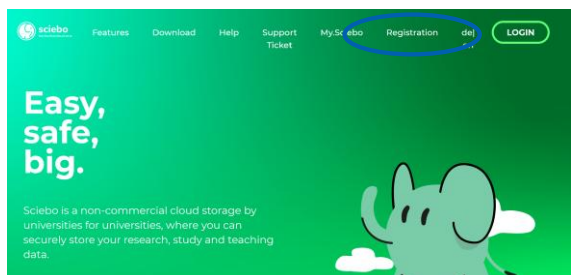
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- <https://hochschulcloud.nrw/en/index.html>



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- Registration / Select University

To register for sciebo or to change your settings, select your institution. You will then be taken to your organisation's login page.

Universität Bonn

Auswählen / Select

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Create Sciebo Account



Login with your university ID and password

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UNIVERSITY IT AND DATA CENTER DE EN

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Login to sciebo - the Campuscloud

Uni-ID
e.g. ab1234567

Password
your password

☐ Don't remember login

☐ Clear prior granting of permission for release of your information to this service

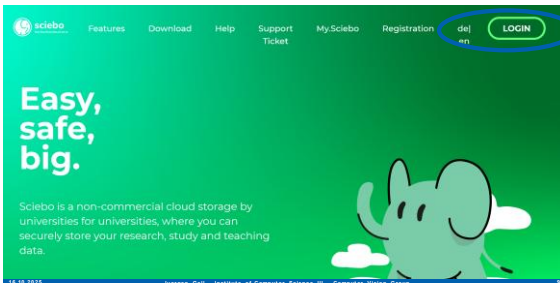
Login

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Login into Sciebo Account



- Login via web (or use desktop client)
- <https://hochschulcloud.nrw/en/index.html>

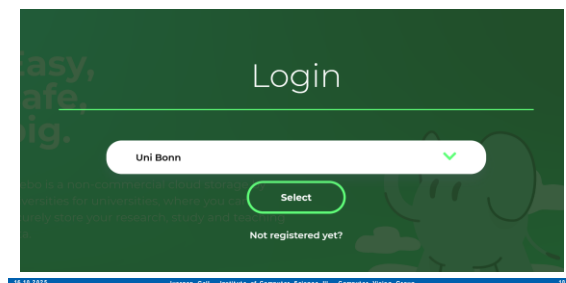


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Login into Sciebo Account



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- Uni email (xyz@uni-bonn.de)
- **Sciebo password** (not university id password)



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Submission of Exercise



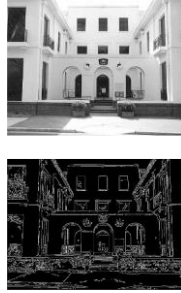
If you have troubles to register or to submit,
please contact
Jinhui Yi
yij@iai.uni-bonn.de

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Overview – Part 1

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- Image
- Image processing
 - Filtering
 - Edges



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Overview – Part 1

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- Detecting objects with single template



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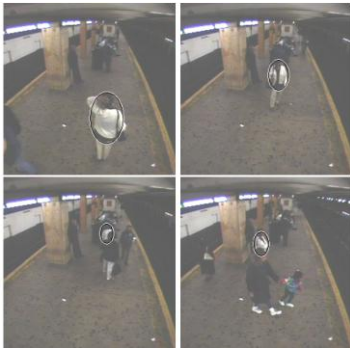
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Overview – Part 1

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- Track objects



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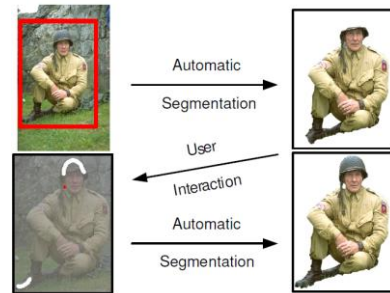
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Overview – Part 1

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- Segment objects



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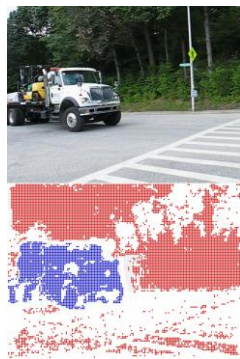
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Overview – Part 1

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- Motion segmentation
- Optical Flow



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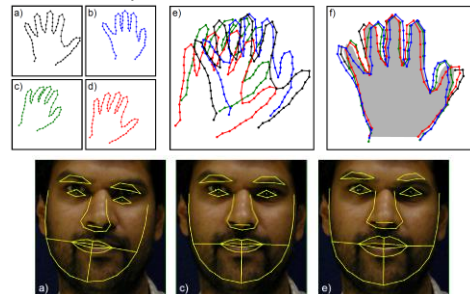
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Overview – Part 1

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- Statistical shape models



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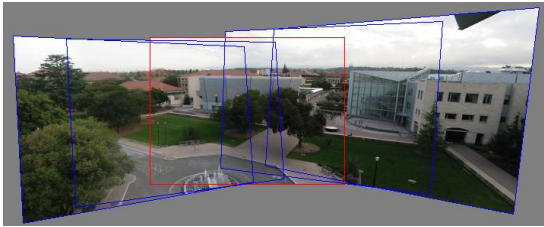
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Overview – Part 1



- Align images / Panorama views



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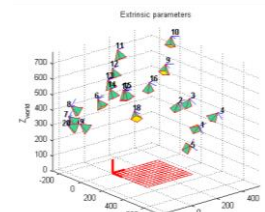
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Overview – Part 2



- Cameras
- Camera calibration



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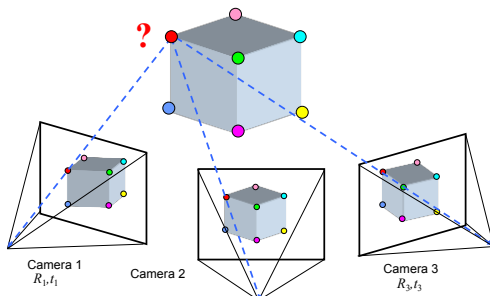
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Overview – Part 2



- 3D Reconstruction / Stereo / Structure from motion



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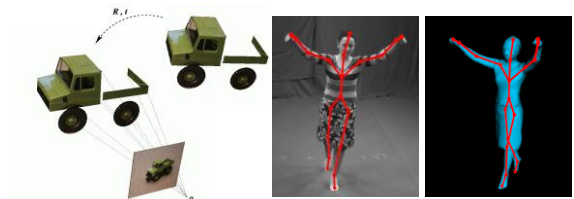
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Overview – Part 2



- Pose estimation



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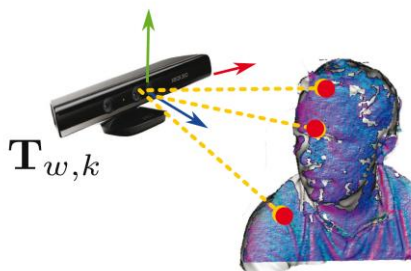
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Overview – Part 2



- Depth sensors

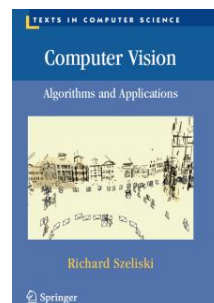


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Literature



Chapter 2 and 3. Computer Vision - Algorithms and Applications, Szeliski, Richard, Springer, 2011

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Image as function



45	60	98	127	132	133	137	133
46	65	98	125	126	128	131	133
47	65	96	115	119	123	135	137
47	65	91	107	112	122	138	124
50	59	80	97	110	125	135	134
49	53	68	83	97	113	128	133
50	50	58	70	84	102	116	126
50	50	52	58	69	86	101	120

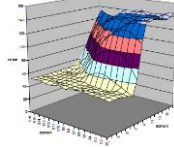
Mapping from image domain (pixels)

to values $\mathbb{R}$

Gray image: $f: \Omega \rightarrow \mathbb{R}$

Color images: $f: \Omega \rightarrow \mathbb{R}^3$

Image represented as matrix or tensor



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Pixel transforms

$$x \in \Omega$$

$$f: \Omega \rightarrow \mathbb{R}$$

Image operator h :

$$g(x) = h(f(x))$$

$$g(i, j) = h(f(i, j)) \quad x = (i, j)$$

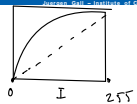
Blend two images linearly

$$g(x) = (1 - \alpha)f_0(x) + \alpha f_1(x)$$

Gamma correction:

$$g(x) = [f(x)]^{1/\gamma}$$

Humans:



$\Rightarrow$ humans can distinguish better in low intensities than in high intensities

Proof ✓

Probability theory:

$$\textcircled{1} \quad \overbrace{F_h(z) = \int_0^z p_h(z') dz'}^{\text{after transformation } h} = \overbrace{\int_0^{h^{-1}(z)} p_f(y') dy'}^{\text{before transformation } h} = F_f(h^{-1}(z))$$

We get:

$$\Rightarrow p_h(z) = \frac{d}{dz} F_h(z) \stackrel{\text{Chain Rule Derivative}}{=} \frac{d}{dz} F_f(h^{-1}(z)) = p_f(h^{-1}(z)) \cdot \frac{d}{dz} h^{-1}(z) \quad (*)$$

Since $h(f(x)) = F(f(x)) \cdot 255$ and $\frac{d}{dy} h(y) = 255 \cdot p_f(y) : (z)$

$$(*) \quad p_f(h^{-1}(z)) \cdot \frac{d}{dz} h^{-1}(z) = \frac{1}{255} \cdot \frac{d}{dy} h(y) \Big|_{y=h^{-1}(z)} \cdot \frac{d}{dz} h^{-1}(z) = \frac{1}{255}$$

$$p_h(z) = \frac{1}{255}$$

$\frac{d}{dz} h(h^{-1}(z)) = 1$
basically identity fct $\Rightarrow 1$

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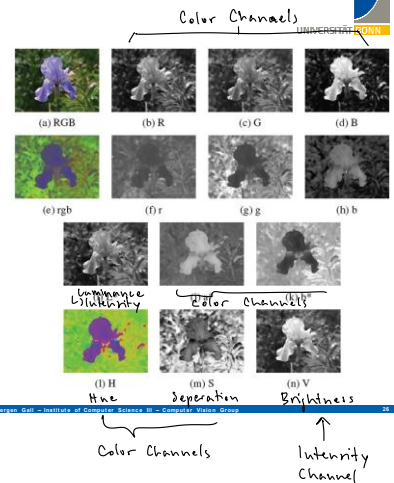
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Color Spaces

There are many color spaces

Popular: Intensity, RGB, Lab, HSV



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disadvantage of rgb:

$\Rightarrow$ no separate channel for image intensity like LAB, HSV

Histogram equalization

Pixels as distribution:

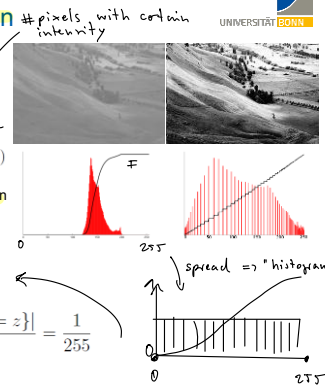
$$p_f(y) = \frac{|\{x \in \Omega : f(x) = y\}|}{|\Omega|} \leftarrow \# \text{pixels in image}$$

$$F_f(y) = \int_0^y p_f(y') dy' = \sum_{y'=0}^y p_f(y')$$

Make CDF linear:

$$h(f(x)) = F(f(x)) \cdot 255$$

$$p_h(z) = \frac{|\{x \in \Omega : h(f(x)) = z\}|}{|\Omega|} = \frac{1}{255}$$

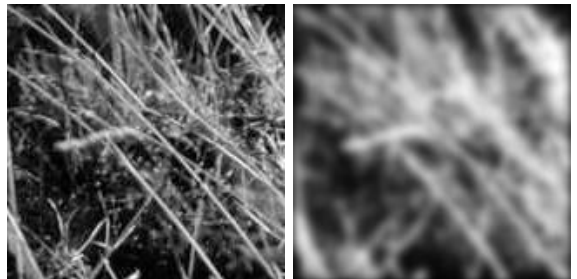


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Linear filtering



Source: S. Lazebnik

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Motivation: Image denoising

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How can we reduce noise in a photograph?



Source: S. Lamboll

Moving average

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Let's replace each pixel with a **weighted average** of its **neighborhood**

The weights are called the filter kernel

What are the weights for the average of a 3x3 neighborhood?

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

"box filter"

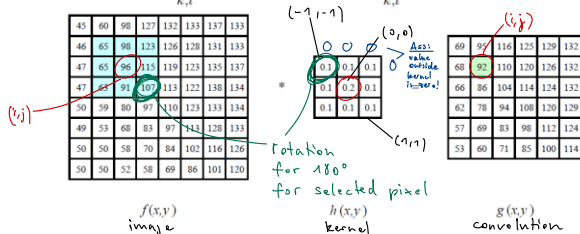
Source: D. Lowe

Defining convolution

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Let f be the image and h be the kernel. The output of convolving f with h is denoted: $g = f * h$

convolution definition: $g(i, j) = \sum_{k, l} f(i - k, j - l) h(k, l) = \sum_{k, l} f(k, l) h(i - k, j - l)$



Source: S. Lamboll

Convolution

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Continuous convolution:

$$g(x) = \int f(x - u) h(u) du$$

h is called impulse response function:

$$h * \delta = h \quad \left(\delta(x) = \begin{cases} \infty & \text{if } x=0 \\ 0 & \text{otherwise} \end{cases} \right)$$

Proof:

$$g(x) = \int \delta(x - u) h(u) du = h(x)$$

Key properties

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- (1) **Linearity:** $\text{filter}(f_1 + f_2) = \text{filter}(f_1) + \text{filter}(f_2)$
 - (2) **Shift invariance:** same behavior regardless of pixel location: $\text{filter}(\text{shift}(f)) = \text{shift}(\text{filter}(f))$
- Theoretical result: any linear shift-invariant operator can be represented as a convolution

given operator:
linear + shift invariant $\Rightarrow$ operator is convolution

Proof

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Linear shift-invariant operator T as a convolution

Using $f(i, j) = \sum_{p=-\infty}^{\infty} \sum_{q=-\infty}^{\infty} f(p, q) \delta(i - p, j - q)$ with $\delta(p, q) = \begin{cases} 1 & \text{if } p=0 \text{ and } q=0 \\ 0 & \text{otherwise} \end{cases}$ we get:

$$(T \circ f)(i, j) = T \left\{ \sum_{p=-\infty}^{\infty} \sum_{q=-\infty}^{\infty} f(p, q) \delta(i - p, j - q) \right\}$$

$$\begin{aligned} & \stackrel{(1)}{=} \sum_{p=-\infty}^{\infty} \sum_{q=-\infty}^{\infty} f(p, q) (T \circ \delta)(i - p, j - q) \quad \text{linear} \\ & \stackrel{(2)}{=} \sum_{p=-\infty}^{\infty} \sum_{q=-\infty}^{\infty} f(i - p', j - q') (T \circ \delta)(p', q') \quad \text{shift-invariant} \\ & = (f * (T \circ \delta))(i, j) \\ & \quad \uparrow \text{convolution!} \end{aligned}$$

Source: S. Lamboll

Source: S. Lamboll

Properties in more detail

Commutative: $a * b = b * a$

- Conceptually no difference between filter and signal

Associative: $a * (b * c) = (a * b) * c$

- Often apply several filters one after another: $((a * b_1) * b_2) * b_3$
- This is equivalent to applying one filter: $a * (b_1 * b_2 * b_3)$

Distributes over addition: $a * (b + c) = (a * b) + (a * c)$

Scalars factor out: $ka * b = a * kb = k(a * b)$

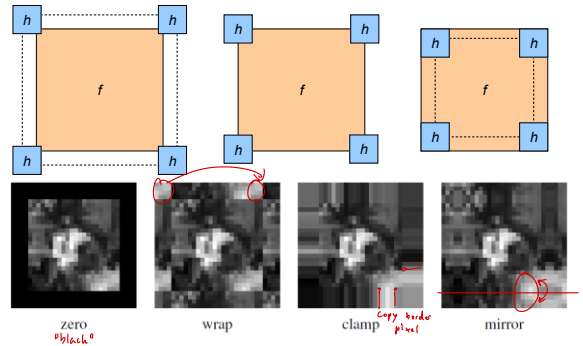
Identity: unit impulse $e = [\dots, 0, 0, 1, 0, 0, \dots]$,
 $a * e = a$

Source: S. Lamberth

Motivation: Convolution would shrink image!

idea: add border before convolution

Border padding and output size

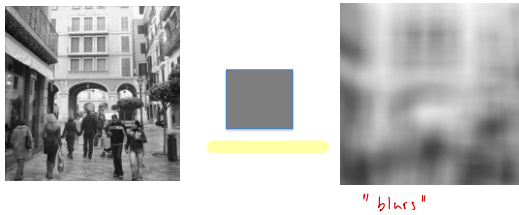


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Rectangular filter



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Source: A. Tomrala

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Rectangular filter



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Source: A. Tomrala

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Rectangular filter



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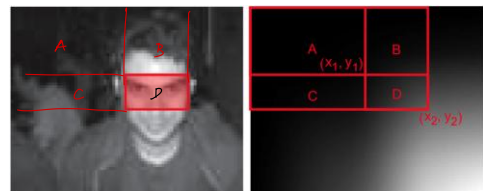
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Source: A. Tomrala

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Integral image

Precompute average



Get average values of any size by reading only 4 values!

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Source: A. Tomrala

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Integral image

3	2	7	2	3
1	5	1	3	4
5	1	3	5	1
4	3	2	1	6
2	4	1	4	8

Original image

3	5	12	14	17
4	11	19	24	31
9	17	28	38	46
13	24	37	48	62
15	30	44	59	81

Integral image

$$s(i, j) = \sum_{k=0}^i \sum_{l=0}^j f(k, l)$$

Recursive:

$$s(i, j) = s(i-1, j) + s(i, j-1) - s(i-1, j-1) + f(i, j)$$

$$17 + 19 - 11 + 3 = 28$$

update
in place

Integral image

3	2	7	2	3
1	5	1	3	4
5	1	3	5	1
4	3	2	1	6
2	4	1	4	8

Original image

3	5	12	14	17
4	11	19	24	31
9	17	28	38	46
13	24	37	48	62
15	30	44	59	81

Integral image

3	5	12	14	17
4	11	19	24	31
9	17	28	38	46
13	24	37	48	62
15	30	44	59	81

Integral image

=> 4 values independent of size:

$$s(i_1, j_1) - s(i_1, j_0 - 1) - s(i_0 - 1, j_1) + s(i_0 - 1, j_0 - 1)$$

$$48 - 13 - 14 + 3 = 24 = 5 + 1 + 3 + 1 + 3 + 5 + 3 + 2 + 1$$

=> calculate pixel sum



Practice with linear filters



Original

0	0	0
0	1	0
0	0	0

?



Original

0	0	0
0	1	0
0	0	0

Identity

Filtered
(no change)

Practice with linear filters



Original

0	0	0
1	0	0
0	0	0

?



Original

0	0	0
1	0	0
0	0	0

Shifted left
By 1 pixel

Practice with linear filters

Remind: Convolution Rotation about 180°

Practice with linear filters



Original

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

?

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Source: D. Lowe

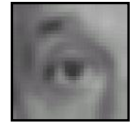
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Practice with linear filters



Original

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$



Blur (with a
box filter)
"smoothing"

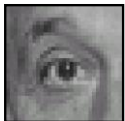
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Source: D. Lowe

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Practice with linear filters



Original

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

(Note that filter sums to 1)

?

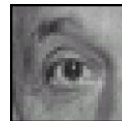
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Source: D. Lowe

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Practice with linear filters



Original

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$



Sharpening filter
- Accentuates differences with local
average

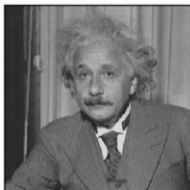
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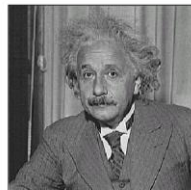
Source: D. Lowe

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Sharpening



before



after

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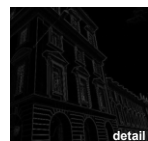
Source: D. Lowe

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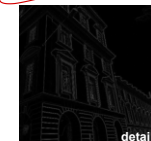
Sharpening



What does blurring take away?



Let's add it back:



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Source: D. Lowe

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Image rotation



It is linear, but not a spatially invariant operation. There is no convolution.

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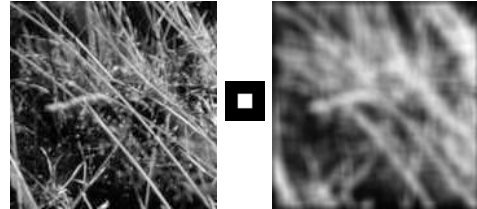
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Source: A. Tomrala

Smoothing with box filter revisited



What's wrong with this picture?



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Source: D. Forsyth

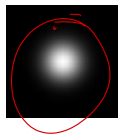
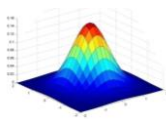
Gaussian Kernel



Constant factor at front makes volume sum to 1 (can be ignored when computing the filter values, as we should renormalize weights to sum to 1 in any case)

$$G_{\sigma} = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

only this relevant



0.003	0.013	0.022	0.013	0.003
0.013	0.059	0.097	0.059	0.013
0.022	0.097	0.159	0.097	0.022
0.013	0.059	0.097	0.059	0.013
0.003	0.013	0.022	0.013	0.003

5 x 5, $\sigma = 1$

Smoothing factor

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Source: C. Rasmussen

Choosing kernel width

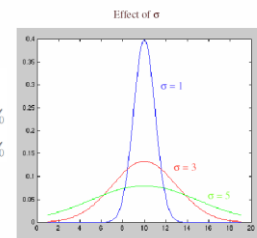


Rule of thumb: set filter half-width to about 3σ

$$P(\mu - \sigma < X \leq \mu + \sigma) \approx 68\%$$

$$P(\mu - 2\sigma < X \leq \mu + 2\sigma) \approx 95.5\%$$

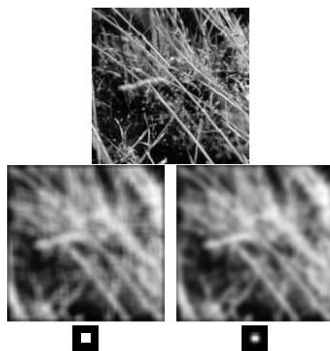
$$P(\mu - 3\sigma < X \leq \mu + 3\sigma) \approx 99.7\%$$



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Gaussian vs. box filtering



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Source: S. Lantieri

Gaussian filters



Remove "high-frequency" components from the image (low-pass filter)

Convolution with itself is another Gaussian

– So can smooth with small- σ kernel, repeat, and get same result as larger- σ kernel would have

– Convoluting two times with Gaussian kernel with std. σ is same as convoluting once with kernel with std. dev. $\sigma\sqrt{2}$

Separable kernel

– Factors into product of two 1D Gaussians

property 1

property 2

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Source: K. Grauman

Separability of the Gaussian filter

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$$G_{\sigma}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

$$= \left(\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right)\right) \left(\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{y^2}{2\sigma^2}\right)\right)$$

The 2D Gaussian can be expressed as the product of two functions, one a function of x and the other a function of y

In this case, the two functions are the (identical) 1D Gaussian

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Separability example

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2D convolution (center location only)

1	2	1
2	4	2
1	2	1

 $*$

2	3	3
3	5	5
4	4	6

The filter factors into a product of 1D filters:

1	2	1
2	4	2
1	2	1

 $=$

1
2
1

 $\times$

1	2	1
---	---	---

Perform convolution along rows:

1	2	1
---	---	---

 $*$

2	3	3
3	5	5
4	4	6

 $=$

11
18
18

Followed by convolution along the remaining column:

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Why is separability useful?

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What is the complexity of filtering an $n \times n$ image with an $m \times m$ kernel?

– $O(n^2 m^2)$

$$\begin{matrix} m \times m \\ \boxed{} \end{matrix} \Rightarrow O(n^2 m^2)$$

What if the kernel is separable?

– $O(n^2 m)$

$$\begin{matrix} m \times m \\ \boxed{} \end{matrix} = \begin{matrix} m \times 1 \\ \boxed{} \end{matrix} + \begin{matrix} 1 \times m \\ \boxed{} \end{matrix} = 2nm = O(n^2 m)$$

How do I know if a kernel is separable?

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Is a kernel separable?

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• Kernel as matrix:

Eigenvalues

$$\text{Kernel matrix } K = \sum_i \sigma_i u_i v_i^T$$

• Singular Value Decomposition (SVD)

• If only the first singular value σ_0 is non-zero, kernel is separable

$$\sigma_0 > 0, \sigma_1 = 0, \dots$$

• Vertical and horizontal kernels:

$$\hookrightarrow \sqrt{\sigma_0} u_0 \text{ and } \sqrt{\sigma_0} v_0^T$$

• Approximation: $K \approx \sigma_0 u_0 v_0^T$

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SVD - Definition

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$$A[m \times n] = U[m \times r] \Sigma[r \times r] (V[n \times r])^T$$

A: Input data matrix

– $m \times n$ matrix

U: Left singular vectors

– $m \times r$ matrix

Σ : Singular values

– $r \times r$ diagonal matrix (r : rank of the matrix A)

V: Right singular vectors

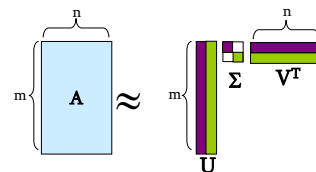
– $n \times r$ matrix

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SVD

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$$A \approx U \Sigma V^T = \sum_i \sigma_i u_i \circ v_i^T$$

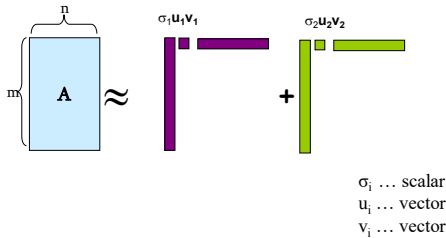


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SVD

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$$A \approx U \Sigma V^T = \sum_i \sigma_i u_i \circ v_i^T$$



Source: J. Leskovec

SVD - Properties

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It is always possible to decompose a real matrix A into $A = U \Sigma V^T$, where

U, Σ, V : unique

U, V : column orthonormal:

- $U^T U = I; V^T V = I$ (I : identity matrix)
- (Cols. are orthogonal unit vectors)

Σ : diagonal

- Entries (singular values) are positive, and sorted in decreasing order ($\sigma_1 \geq \sigma_2 \geq \dots \geq 0$)

$$u_1^T u_2 = 0$$

$$u_1^T u_1 = 1$$

Source: J. Leskovec

SVD – Example

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$A = U \Sigma V^T$ - example:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.18 & 0 \\ 0.36 & 0 \\ 0.18 & 0 \\ 0.90 & 0 \\ 0 & 0.53 \\ 0 & 0.80 \\ 0 & 0.27 \end{bmatrix} \times \begin{bmatrix} 9.64 & 0 \\ 0 & 5.29 \end{bmatrix} \times \begin{bmatrix} 0.58 & 0.58 & 0.58 & 0 & 0 \\ 0 & 0 & 0 & 0.71 & 0.71 \end{bmatrix}$$

Eigenvalues (pointing to the diagonal of Σ)

rank = 2 (under the first two columns of U)

U (under the first matrix)

V (under the third matrix)

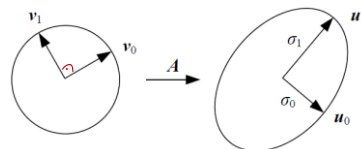
rank = #linear independent columns

Source: J. Leskovec

Geometric interpretation

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Change of basis vectors:



$$A_{M \times N} = U_{M \times P} \Sigma_{P \times P} V_{P \times N}^T$$

$$AV = U\Sigma \quad \text{or} \quad Av_j = \sigma_j u_j$$

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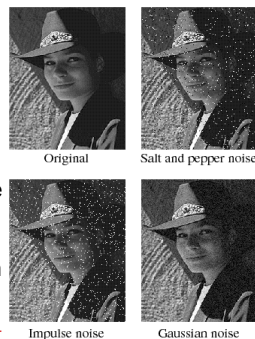
Noise

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Salt and pepper noise: contains random occurrences of black and white pixels

Impulse noise: contains random occurrences of white pixels

Gaussian noise: variations in intensity drawn from a Gaussian normal distribution



Source: S. Seitz

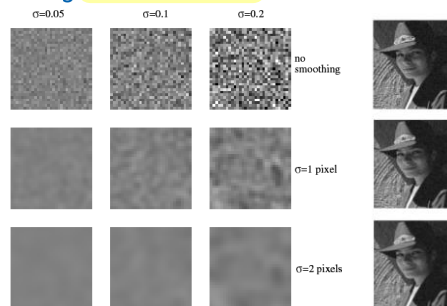
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Reducing Gaussian noise

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Smoothing with larger standard deviations suppresses noise, but also blurs the image

Source: S. Seitz

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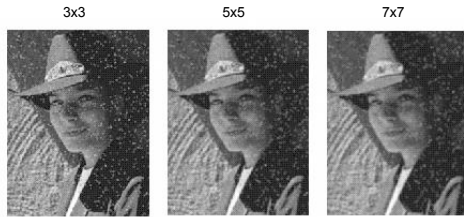
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Reducing salt-and-pepper noise

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What's wrong with the results?



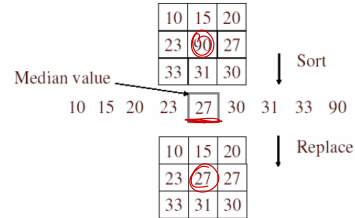
→ Gaussian filter does not work for Salt + Pepper Noise
→ We Median filter instead

Source: S. Lazebnik

Alternative idea: Median filtering

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A median filter operates over a window by selecting the median intensity in the window



Is median filtering linear?

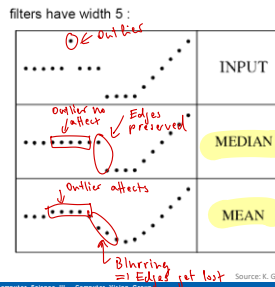
Source: K. Grauman

Median filter

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What advantage does median filtering have over Gaussian filtering?

Robustness to outliers



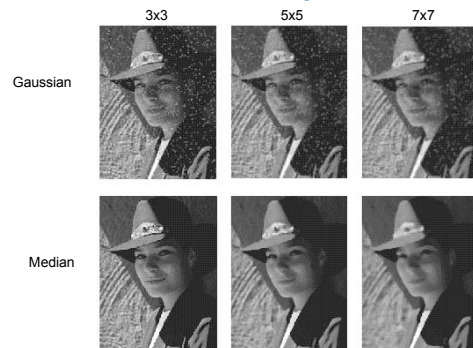
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Source: K. Grauman

Gaussian vs. median filtering

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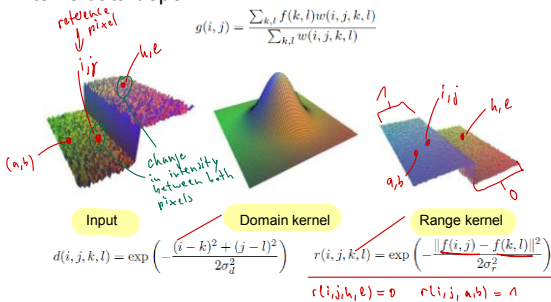
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Source: S. Lazebnik

Bilateral filter

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Filter is data dependent



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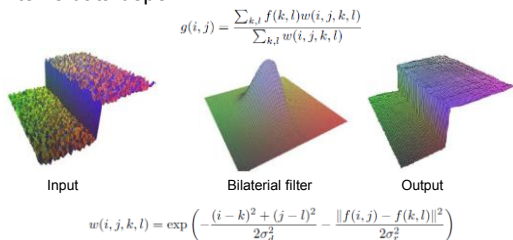
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Bilateral filter

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Filter is data dependent



Slow, but methods for approximation exist

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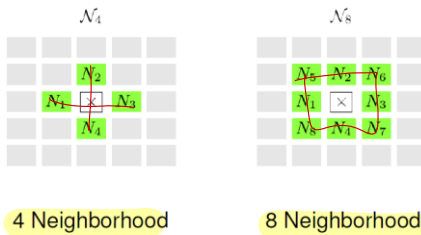
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Binary images

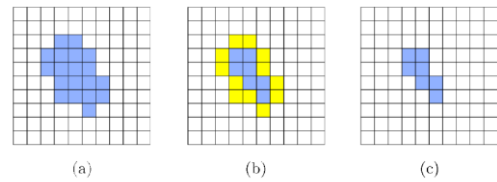


Neighborhoods



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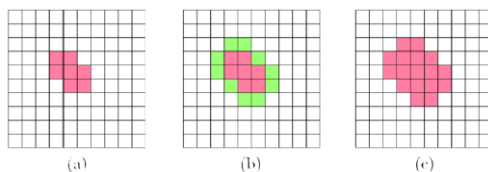
Erosion



Change a foreground pixel to background if it has a background pixel as a 4-neighbor.

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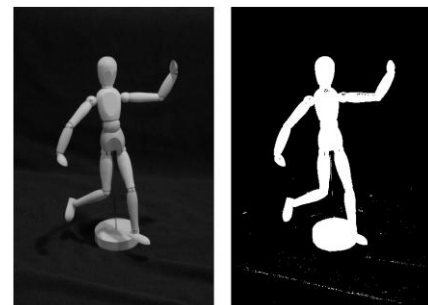
Dilation



Change a background pixel to foreground if it has a foreground pixel as a 4-neighbor.

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Threshold

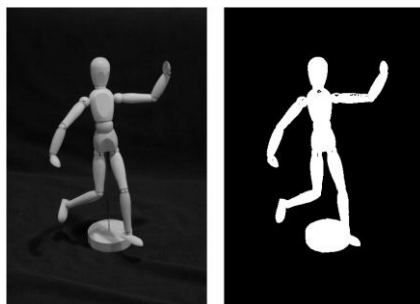


Original image

Initial threshold

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Opening = Dilate(Erode)

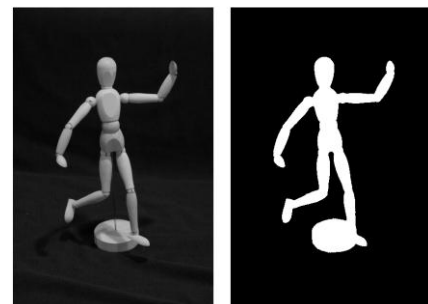


Original image

After opening

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Closing = Erode(Dilate)



Original image

After closing

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Thank you for your attention.

